

Python Programming Assignment

Name: UDHAYAKUMAR U
Register Number: 192424332

Question 36(a): Manufacturing Production Monitoring (10 Marks)

A manufacturing company stores production records in CSV format and machine details in XML format. Write a Python program to identify machines with the highest production output and generate a machine performance report.

Sample Input: machines.xml

```
<Machines>
  <Machine>
    <MachineID>MC1</MachineID>
    <Name>CNC Lathe A1</Name>
    <Section>Turning</Section>
  </Machine>
  <Machine>
    <MachineID>MC2</MachineID>
    <Name>Press Machine B2</Name>
    <Section>Stamping</Section>
  </Machine>
  <Machine>
    <MachineID>MC3</MachineID>
    <Name>Welding Robot C3</Name>
    <Section>Assembly</Section>
  </Machine>
</Machines>
```

Sample Input: production.csv

```
RecordID,MachineID,Date,UnitsProduced
PR1,MC1,2026-07-01,450
PR2,MC2,2026-07-01,600
PR3,MC3,2026-07-01,320
PR4,MC1,2026-07-02,470
PR5,MC2,2026-07-02,580
PR6,MC3,2026-07-02,340
```

Python Program

```
import xml.etree.ElementTree as ET
import csv

# Read machine details from XML
machines = {}
tree = ET.parse('machines.xml')
for m in tree.getroot().findall('Machine'):
    mid = m.find('MachineID').text
    machines[mid] = {'Name': m.find('Name').text, 'Section': m.find('Section').text}

# Read production records from CSV, total units per machine
totals = {mid: 0 for mid in machines}
with open('production.csv') as f:
    for row in csv.DictReader(f):
        totals[row['MachineID']] += int(row['UnitsProduced'])

# Sort machines by total output, descending
ranked = sorted(totals.items(), key=lambda x: -x[1])
highest_output = ranked[0][1]

# Generate report
```

```

report = ["MACHINE PERFORMANCE REPORT", f"{'ID':<6}{'Name':<20}{'Section':<12}{'Total Output'}"]
for mid, total in ranked:
    tag = " <-- Highest" if total == highest_output else ""
    report.append(f"{'mid':<6}{'machines[mid]':<20}{'machines[mid]':<12}{total}{tag}")

report_text = "\n".join(report)
print(report_text)
with open('machine_report.txt', 'w') as f:
    f.write(report_text)

```

Sample Output

```

MACHINE PERFORMANCE REPORT
ID      Name                Section      Total Output
MC2     Press Machine B2        Stamping     1180 <-- Highest
MC1     CNC Lathe A1            Turning       920
MC3     Welding Robot C3        Assembly     660

```

Question 36(b): Hospital Vaccination Management (10 Marks)

A hospital stores patient information in JSON format and vaccination records in CSV format. Write a Python program to identify patients who are due for their next vaccination and generate reminder reports.

Sample Input: patients.json

```

[
    {"PatientID": "P1", "Name": "Ramesh Iyer", "Age": 32},
    {"PatientID": "P2", "Name": "Lakshmi Devi", "Age": 45},
    {"PatientID": "P3", "Name": "Karthik S", "Age": 27},
    {"PatientID": "P4", "Name": "Anjali Menon", "Age": 60}
]

```

Sample Input: vaccinations.csv

```

RecordID,PatientID,VaccineName,DoseNumber,DateGiven,NextDueDate
V1,P1,Hepatitis B,1,2026-06-01,2026-07-01
V2,P2,Hepatitis B,1,2026-07-10,2026-08-10
V3,P3,Tetanus,1,2026-05-15,2026-07-15
V4,P4,Influenza,1,2026-01-10,2027-01-10

```

Python Program

```

import json
import csv
from datetime import date

TODAY = date(2026, 7, 25) # reference date for checking due vaccinations

# Read patient details from JSON
with open('patients.json') as f:
    patients = {p['PatientID']: p for p in json.load(f)}

# Read vaccination records from CSV
due_list = []
with open('vaccinations.csv') as f:
    for row in csv.DictReader(f):
        next_due = date.fromisoformat(row['NextDueDate'])
        if next_due <= TODAY:
            due_list.append((row['PatientID'], row['VaccineName'], next_due))

# Generate reminder report
report = ["VACCINATION REMINDER REPORT", f"{'ID':<6}{'Name':<15}{'Vaccine':<15}{'Due Date'}"]
for pid, vaccine, due_date in due_list:
    name = patients[pid]['Name']
    report.append(f"{'pid':<6}{'name':<15}{'vaccine':<15}{'due_date'}")

report.append(f"\nTotal patients due for vaccination: {len(due_list)}")

```

```
report_text = "\n".join(report)
print(report_text)
with open('vaccination_report.txt', 'w') as f:
    f.write(report_text)
```

Sample Output

```
VACCINATION REMINDER REPORT
ID      Name      Vaccine      Due Date
P1      Ramesh Iyer    Hepatitis B   2026-07-01
P3      Karthik S      Tetanus      2026-07-15
```

```
Total patients due for vaccination: 2
```